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Project description:

Urdu C is a mini-programming language inspired by C and C++ that uses Urdu-based keywords instead of English ones. The main goal of this language is to make programming easy and understandable for beginners. The syntax is same as of C/C++.

Regex Table:

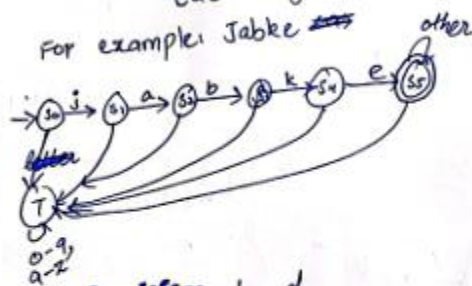
Token Type	Example	Regular expression	Explanation
Identifier	Number, age	<code>[a-zA-Z_][a-zA-Z0-9_]*</code>	Any English word and alphabet can be used
Integer Literal	10, 256, 0	<code>0 [1-9][0-9]*</code>	Any number can be used
Float Literal	12.34, 0.56, 99.0	<code>([0-9]+(\.[0-9]*)?)</code>	Any decimal number can be used
Character Literal	'A', '9'	<code>'(?:[^ ' \] \\.)'</code>	Any alphabet in single quotation can be used
String Literal	"Hello", "Hi"	<code>"(?:[^ " \] \\.)**"</code>	Any alphabet in double quotation can be used
Keywords	Jabtak (a > 0){}	jabke tophir jabtak harbar lautao khali haan batil taaza mitao दौरا likhye lijiye aurphir yan	Keywords specified can be used
Data Types	int num ;	int float bool char string khali	Any data type which is specified is used
Operators	Num ++	+ ++ -- = = = < > <= >= && / \\ + * % !	Operators like these can be used

State diagrams:

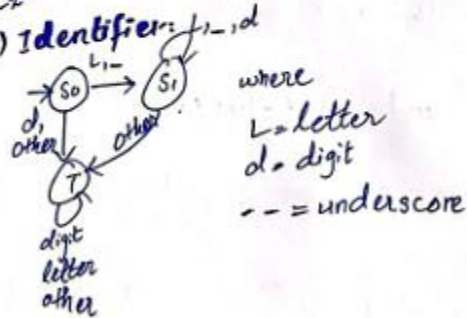
State Diagram:

i) keyword:

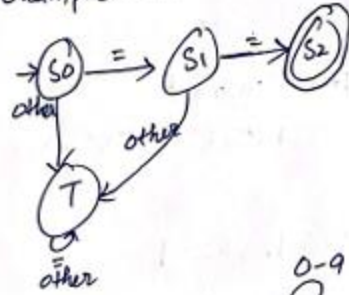
Each keyword has its own diagram.
For example: Jabke ~~xxxx~~



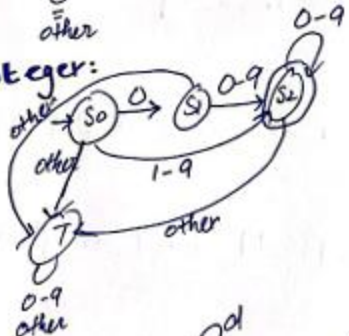
ii) Identifier: $\{L, _, d\}^*$



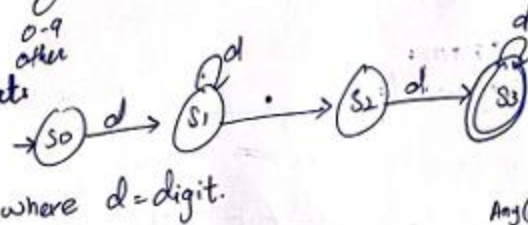
iii) Operators: Every operator has its own diagram.
Example (=)



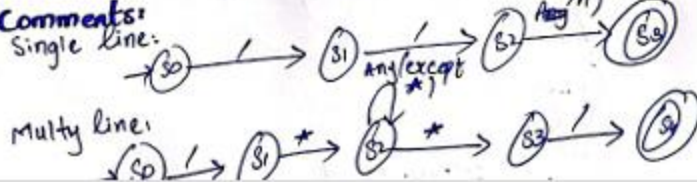
iv) Integer:



(v) Floats:



(v) Comments:



Explanation:

Keywords like jabtak, jabke, and tohphir because it is easy to understand Urdu language for Pakistani programmers and it also makes programmers easy to understand the concept of these keywords and use them in programming.